As you can see, consensus is a core component of a blockchain protocol, and an efficient algorithm is essential for both the integrity and scalability of blockchain. Scalability is the next challenge I'll discuss.

12.4 Scalability

Scalability is a bottleneck preventing the broader adoption of blockchain for business applications. The question many businesses ask is whether blockchain protocol, infrastructure, network, and nodes can successfully deliver a transaction rate on par with that of credit card transactions. Average transaction confirmation time depends on the average block time--the time required to confirm a block. As shown in figure 12.3, transactions are packed in a block, and the block is appended to the chain. All the transactions in a confirmed block have the timestamp of the block. You must be aware of this fact when using transaction confirmation times for verification in your application.

> **DEFINITION** *Scalability* is the ability of a system to perform satisfactorily at all practical levels of load. Load, in the context of the blockchain, could be transaction times, transaction rate, number of nodes, number of participants and accounts, number of transactions, or other attributes.

In the case of blockchain, practitioners are concerned about the transaction rate, or transactions per second. This metric is critical for many applications, from payment systems to supply chain management, so let's focus on transactions per second as the metric for scalability.

Blockchain has taken on the responsibilities of the intermediaries, including validation, verification, and recording of the transaction. The consensus process for the integrity of the chain is another time-consuming function. All these functions